

Debt & Loans

ProfitSync · engineering report · branch `feat/debt-loans` (stacked on `feat/credit-card-accounts` , PR #365) · 4 September 2026

Debts and loans are now first-class in the ledger. A debt is a **liability account**; money someone owes the user is a **receivable** account. Borrowing is a transfer into a bank or cash account, never income. Repaying principal is a transfer back, never an expense. Only interest, fees and other charges are expenses. Everything the hub shows — status, progress, this month's obligations, next payment, debt-free date, payoff plans — is derived from the ledger plus the terms the user entered. No parallel bookkeeping was added.

793

unit tests green (68 files,
+58 new)

11 / 11

end-to-end debt
scenarios on a real
database

19 / 19

credit-card + smoke e2e
regression still green

1

migration (0063), additive,
legacy rows untouched

1. What I found (audit)

- `wealth_accounts.current_balance` already holds the **signed asset-equivalent** value for every account type; credit cards (PR #365) are negative when owed. The ledger (`balanceDelta` , transfers, trash/restore reversals, system rows) needed no change to host a loan.
- Loans had no representation. The only way to record one was "income" on borrowing and "expense" on every instalment, which inflated both sides of every report.
- The plain transaction endpoints accepted any account id, so a loan account would have been spendable like a bank. Net worth only knew liquid accounts and cards.
- Multi-currency: accounts carry no currency of their own, so a foreign-currency loan needs a native currency on its terms and totals must stay per currency.

2. Architecture — a debt IS a wealth account

A debt is a `wealth_accounts` row with `type = 'loan'` (I owe, balance negative) or `'receivable'` (owed to me, balance positive). Its TERMS live in a 1:1 `debt_details` row. Every movement is an ordinary ledger row, so transfers, trash, restore, purge, cascades, budgets and net worth work unchanged. The sign is read only through `isLiabilityType` / `src/lib/debt-status.ts` , never in a component.

Event	Ledger rows	Debt	Bank	Expense	Income	Net worth
Borrow €5,000 into cash	transfer loan → cash (or a system Opening Balance when the money is already spent)	−5,000	+5,000	0	0	0
Pay €500: 420 principal, 70 interest, 10 fees	ONE group: transfer bank → loan €420 + two outgoing expenses on the bank	+420	−500	+80	0	−80
Lend €400 to Luca	transfer cash → receivable	+400 (asset)	−400	0	0	0
Luca repays €100 + €5 interest	transfer receivable → cash €100 + incoming €5 on cash	−100	+105	0	+5	+5
Reconcile balance to a statement	system Balance Adjustment on the debt	Δ	—	0	0	Δ

Terms vs derived facts

`debt_details` stores only what the user knows: kind, counterparty, native currency, original amount, rate and rate type, scheduled payment and frequency, next due date, start/maturity, remaining instalments, an *estimate* flag, the lifecycle (`active` | `paused` | `paid_off` | `refinanced` | `written_off`), the refinanced-into account and notes. Status, progress, this month's required/paid/remaining, overdue, next payment, remaining interest, the debt-free date and the insights are all computed in `src/lib/debt-status.ts` from the balance and the payment history. Nothing that can be derived is stored.

Payments and their split

A recorded repayment is one ledger **group**: the principal transfer plus zero or more expense legs for interest, fees and other charges on the paying account. `debt_payments` records the allocation (`principal` / `interest` / `fees` / `other` , `split_source` ∈ `calculated` | `principal_only` | `typed`) anchored on the group's first leg with a cascading FK, so trashing the group hides the allocation and restoring brings it back once. When the user knows only the total, the split is calculated from one period of interest at the entered rate; with no rate the whole amount is principal and marked as such. Recording a payment also advances `next_due_date` and `remaining_installments` .

Engine (integer cents)

`src/lib/debt-math.ts` : periodic rate per frequency, one-period interest, annuity payment, amortization with a final-payment tolerance (`min(payment, max(€1, 2 %))`) so schedules converge without a phantom extra period, and a non-convergence flag when the payment does not cover interest. `src/lib/debt-planner.ts` : month-by-month simulation of **minimum only** (no pool, no rollover), **avalanche** (highest rate first), **snowball** (smallest balance first), **free-up-cash** (highest payment-to-balance first) and **custom order**, with a cleared debt's payment rolling into the next target, what-if extra per month and a one-off lump sum, side-by-side comparison, an affordability check that switches to a stabilisation message when scheduled payments exceed the budget, and a debt-payment-to-income ratio. Debts without a rate are simulated at 0 % and labelled as such.

3. Data model

Migration **0063** (hand-written like 0060–0062; `drizzle-kit generate` is out of sync), purely additive:

- `wealth_accounts.type` accepts `loan` and `receivable` (existing rows untouched).
- `debt_details`: one row per debt account (unique `wealth_account_id`, cascades), CHECK constraints on kind, rate type, frequency, lifecycle and split source, non-negative amounts.
- `debt_payments`: `transaction_id` FK cascade, `group_id`, `date`, `total`, `principal`, `interest`, `fees`, `other`, `split_source`, `note`; read only while the anchor transaction is not trashed.

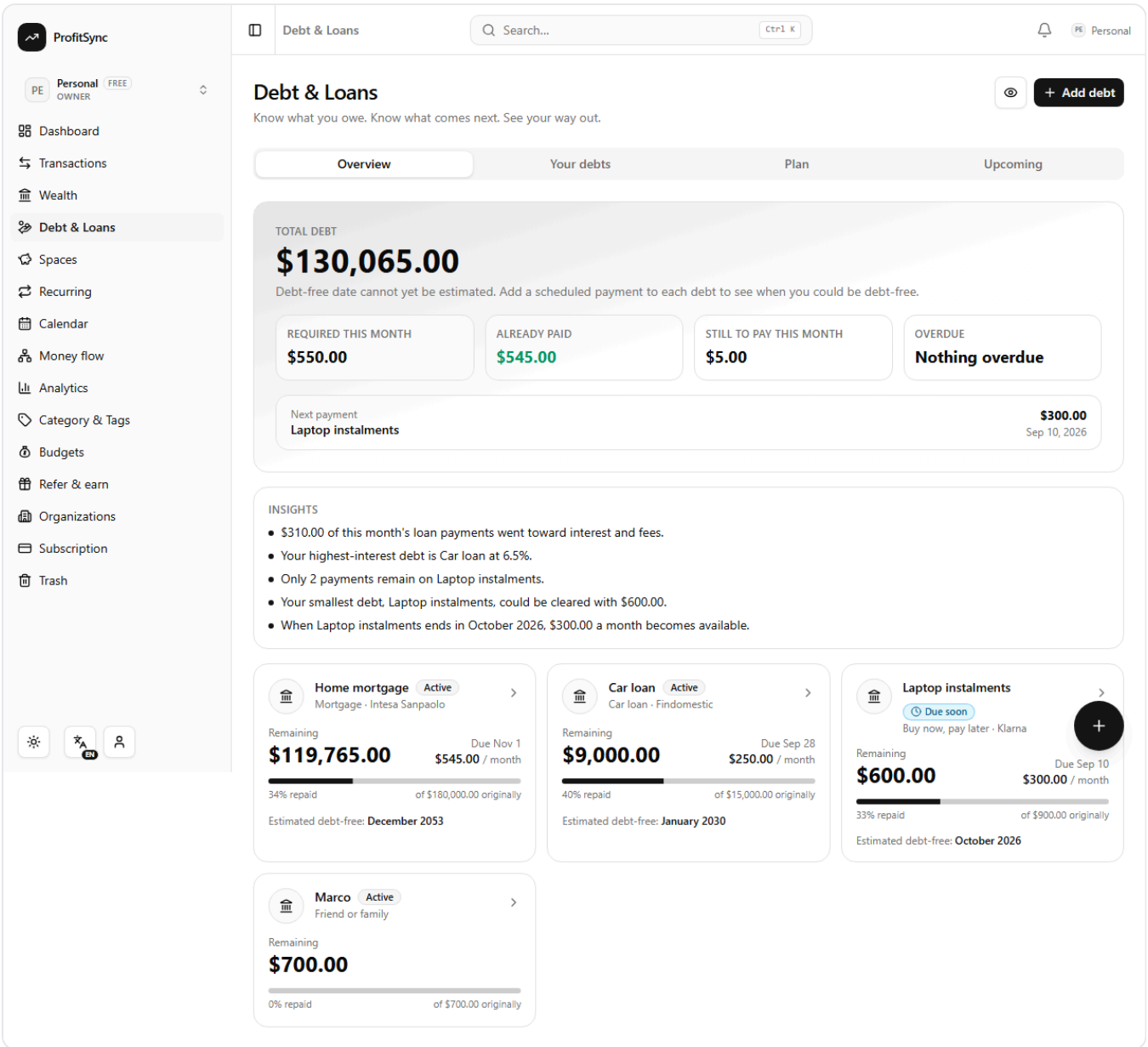
Deleting a debt: an account with history is archived (closed) so its ledger stays consistent; an account with no non-system rows is hard-deleted and its rows cascade. Account deletion / factory reset already cascade `wealth_accounts`, so the new tables follow.

4. API

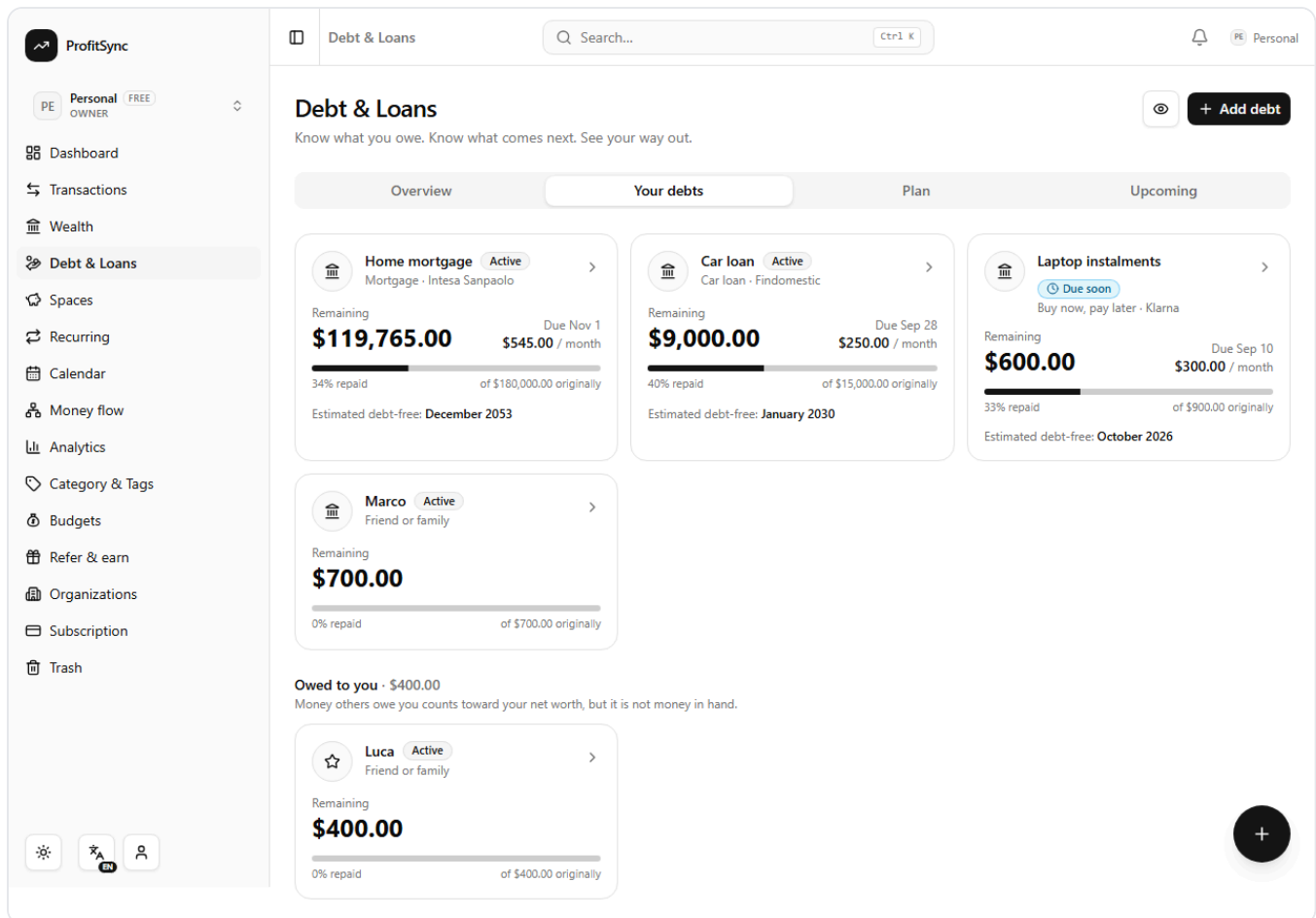
Route	Purpose
<code>GET /api/debts</code>	Hub payload: debts (owed and receivable), summary (per-currency totals, this month required / paid / remaining / overdue, next payment, interest this month, total repaid, debt-free date, average monthly income, insights) and the next three months of scheduled payments.
<code>POST /api/debts</code>	Create (Quick or Detailed). <code>direction</code> = owed / receivable; opening either as a system Opening Balance or as a transfer with <code>disbursement_account_id</code> .
<code>GET / PATCH / DELETE /api/debts/:id</code>	Detail + payment history + projected schedule · edit terms, lifecycle, refinanced-into, reconcile the balance (system Balance Adjustment) · archive-or-delete.
<code>GET / POST /api/debts/:id/payments</code>	List · record a repayment (one ledger group, quota-checked, categories for interest / fees resolved from the org's list).
<code>DELETE /api/debts/:id/payments/:paymentId</code>	Move the whole payment group to Trash (restore via the existing Trash flow).

Guard-rails: `GET /api/wealth/accounts` hides `space`, `loan` and `receivable`; the plain transaction endpoints reject debt accounts; every route is org-scoped through `requireAuth` and role-checked with `canWrite` / `canDelete`. Registered in `api/index.ts` after the card route (static before dynamic).

5. The built UI



/debts — Overview: total debt, required / paid / still to pay this month, overdue, next payment, plain-language insights, and a card per debt with remaining, scheduled payment, due date, progress and an estimated debt-free month where one can be computed.



Your debts — liabilities first, then "Owed to you" as a separate group with the reminder that it counts toward net worth but is not money in hand.

ProfitSync

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Dashboard

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Wealth

Debt & Loans

Spaces

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Money flow

Analytics

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Budgets

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Trash

Debt & Loans

Search...Ctrl K

Personal

Debt & Loans

Know what you owe. Know what comes next. See your way out.

OverviewYour debtsPlanUpcoming

Choose how much you can put toward debt each month and compare ways to clear it. Based on the balances and rates you entered.

Scheduled payments\$1,095.00/month

Monthly debt budget1095

Scheduled payments are \$1,095.00. Anything above that is extra repayment.

Minimum onlyKeep paying exactly what is scheduled today.

Save the most interestExtra money goes to the highest interest rate first.

Fastest early winsExtra money goes to the smallest balance first.

Free up monthly cashExtra money goes where it frees the most monthly cash.

My own orderYou decide which debt comes first.

SAVE THE MOST INTEREST

August 2038

REMAINING INTEREST\$25,566.95

FIRST DEBT CLEAREDLaptop instalments · 2 months

ESTIMATEDassumes 0 % (rate not entered)

Projected balance

Today\$130,065.00

September 2029\$101,729.00

September 2032\$70,306.75

September 2035\$35,826.16

August 2038\$

When a debt is cleared, its payment rolls into the next one automatically. Estimates are based on the balances, payments and rates you entered.

What if...

Extra per month0

One-off payment today0

Strategy	Debt-free	Remaining interest	First debt cleared
Minimum only	not within 50 years	\$58,423.77	Laptop instalments · 2 months
Save the most interest	August 2038	\$25,566.95	Laptop instalments · 2 months
Fastest early wins	August 2038	\$25,897.93	Laptop instalments · 2 months
Free up monthly cash	August 2038	\$25,566.95	Laptop instalments · 2 months
My own order	September 2038	\$25,993.69	Laptop instalments · 2 months

Save the most interest is estimated to save the most interest.

Plan — monthly debt budget (input + slider), five strategies as radio cards, the selected plan's debt-free month, remaining interest, first debt cleared, the "assumes 0 %" marker for debts without a rate, a projected-balance chart, what-if inputs and the comparison table.

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Debt & Loans

Q Search...Ctrl K

🔔PEPersonal

Debt & Loans

Know what you owe. Know what comes next. See your way out.

👁️

+ Add debt

Overview

Your debts

Plan

Upcoming

September 2026 debt payments

\$550.00 · Paid \$0.00 · Remaining \$550.00

Sep 10Laptop instalments\$300.00

Sep 28Car loan\$250.00

October 2026 debt payments

\$550.00 · Paid \$0.00 · Remaining \$550.00

Oct 10Laptop instalments\$300.00

Oct 28Car loan\$250.00

November 2026 debt payments

\$1,095.00 · Paid \$0.00 · Remaining \$1,095.00

Nov 1Home mortgage\$545.00

Nov 10Laptop instalments\$300.00

Nov 28Car loan\$250.00

+

Upcoming — scheduled payments grouped by month with required / paid / remaining per month.

Personal

OWNER

FREE

Debt & Loans

Search...

Ctrl K

Personal

Home mortgage

Active

Mortgage · Intesa Sanpaolo

REMAINING

\$119,765.00

34% repaid · of \$180,000.00 originally

PAYMENT

\$545.00 every month

NEXT DUE

Nov 1, 2026

RATE

3.1%

ESTIMATED DEBT-FREE

December 2053

Estimated remaining interest: \$57,385.26

+ Record payment

Edit

Payment history (1)

\$545.00

Sep 4, 2026 · Principal \$235.00 · Interest \$310.00

Schedule

Based on the balance, payment and rate entered.

Date	Payment	Interest	Principal	Balance
Nov 1, 2026	\$545.00	\$309.39	\$235.61	\$119,529.39
Dec 1, 2026	\$545.00	\$308.78	\$236.22	\$119,293.17
Jan 1, 2027	\$545.00	\$308.17	\$236.83	\$119,056.34
Feb 1, 2027	\$545.00	\$307.56	\$237.44	\$118,818.90
Mar 1, 2027	\$545.00	\$306.95	\$238.05	\$118,580.85
Apr 1, 2027	\$545.00	\$306.33	\$238.67	\$118,342.18

Show full schedule

/debts/:id — remaining, progress, payment / next due / rate / estimated debt-free, remaining interest, Record payment and Edit, payment history with the split, and the projected schedule (interest, principal, balance per period).

Personal

OWNER

FREE

Debt & Loans

Search...

Ctrl K

Personal

Debt & Loans

Know what you owe

OVERVIEW

Nothing overdue

ADD DEBT

I owe

Owed to me

Quick

Detailed

Who do you owe?

Bank, shop or a person — e.g. Intesa, Marco

Name (optional)

Type

How much is left?

Currency

This is an estimate

every

What do you normally pay?

Monthly

When is the next payment?

dd/mm/yyyy

Interest rate — leave empty if none or unknown

0.00

Cancel

Add

Personal

OWNER

FREE

Debt & Loans

Search...

Ctrl K

Personal

Car loan

Active

Car loan · Intesa Sanpaolo

REMAINING

\$9,000.00

34% repaid · of \$180,000.00 originally

PAYMENT

\$250.00 every month

NEXT DUE

Nov 1, 2026

RATE

3.1%

ESTIMATED DEBT-FREE

January 2030

Estimated remaining interest: \$57,385.26

+ Record payment

Edit

Payment history (1)

\$250.00

Sep 4, 2026 · Principal \$235.00 · Interest \$310.00

Schedule

Based on the balance, payment and rate entered.

Date	Payment	Interest	Principal	Balance
Sep 28, 2026	\$250.00	\$40.75	\$209.25	\$8,790.75
Oct 28, 2026	\$250.00	\$40.66	\$209.34	\$8,596.41
Nov 28, 2026	\$250.00	\$40.56	\$209.44	\$8,401.97
Dec 28, 2026	\$250.00	\$40.46	\$209.54	\$8,207.43
Jan 28, 2027	\$250.00	\$40.36	\$209.64	\$8,012.79

Show full schedule

Add debt (I owe / Owed to me, Quick / Detailed, estimate toggle, native currency, "leave empty if none or unknown" rate) and Record payment ("Split automatically" preview vs "I know the split").

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Debt & Loans

+ Add debt

Know what you owe. Know what comes next. See your way out.

Overview

Your debts

Plan

Upcoming

TOTAL DEBT

\$130,065.00

Debt-free date cannot yet be estimated. Add a scheduled payment to each debt to see when you could be debt-free.

REQUIRED THIS MONTH

\$550.00

ALREADY PAID

\$545.00

STILL TO PAY THIS MONTH

\$5.00

OVERDUE

Nothing overdue

Next payment

Laptop instalments

\$300.00

Sep 10, 2026

INSIGHTS

\$310.00 of this month's loan payments went toward interest and fees.

Your highest-interest debt is Car loan at 6.5%

Only 2 payments remain on Laptop instalments

Your smallest debt, Laptop instalments, could be cleared with \$600.00.

Home

Transactions

More

Home mortgage

Active

Mortgage · Intesa Sanpaolo

Remaining

\$119,765.00

Due Nov 1

\$545.00 / month

34% repaid

of \$180,000.00 originally

Estimated debt-free: December 2053

Car loan

Active

Car loan · Findomestic

Remaining

\$9,000.00

Due Sep 28

\$250.00 / month

40% repaid

of \$15,000.00 originally

Estimated debt-free: January 2030

Laptop instalments

Due soon

Buy now, pay later · Klarna

Remaining

\$600.00

Due Sep 10

\$300.00 / month

33% repaid

of \$900.00 originally

Estimated debt-free: October 2026

Marco

Active

Friend or family

Remaining

\$700.00

0% repaid

of \$700.00 originally

ProfitSync

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Debt & Loans

+ Add debt

Know what you owe. Know what comes next. See your way out.

Overview

Your debts

Plan

Upcoming

Choose how much you can put toward debt each month and compare ways to clear it. Based on the balances and rates you entered.

Scheduled payments

\$1,095.00 / month

Monthly debt budget

1095

Scheduled payments are \$1,095.00. Anything above that is extra repayment.

Minimum only

Keep paying exactly what is scheduled today.

Save the most interest

Extra money goes to the highest interest rate first.

Fastest early wins

Extra money goes to the smallest balance first.

Free up monthly cash

Extra money goes where it frees the most monthly cash.

My own order

SAVE THE MOST INTEREST

August 2038

REMAINING INTEREST

\$25,566.95

FIRST DEBT CLEARED

Laptop instalments · 2 months

ESTIMATED

assumes 0 % (rate not entered)

Projected balance

Today

September 2029

September 2032

September 2035

August 2038

\$130,065.00

\$101,729.00

\$70,306.75

\$35,826.16

\$0.00

When a debt is cleared, its payment rolls into the next one automatically. Estimates are based on the balances, payments and rates you entered.

What if...

Extra per month

0

One-off payment today

0

Strategy

Debt-free

Remain inter

Minimum only

not within 50 years

\$58,423

Save the most interest

August 2038

\$25,566

Fastest early wins

August 2038

\$25,897

Free up monthly cash

August 2038

\$25,566

My own order

September 2038

\$25,993

ProfitSync

PEPersonalFREE

Home mortgage

Active

Mortgage · Intesa Sanpaolo

REMAINING

\$119,765.00

34% repaid · of \$180,000.00 originally

PAYMENT

\$545.00 every month

NEXT DUE

Nov 1, 2026

RATE

3.1%

ESTIMATED DEBT-FREE

December 2053

Estimated remaining interest: \$57,385.26

+ Record payment

Edit

Payment history (1)

\$545.00

Sep 4, 2026 · Principal \$235.00 · Interest \$310.00

Schedule

Based on the balance, payment and rate entered.

Date

Payment

Interest

Principal

Nov 1, 2026

\$545.00

\$309.39

\$235.61

Dec 1, 2026

\$545.00

\$308.78

\$236.22

Jan 1, 2027

\$545.00

\$308.17

\$236.83

Apr 1, 2027

\$545.00

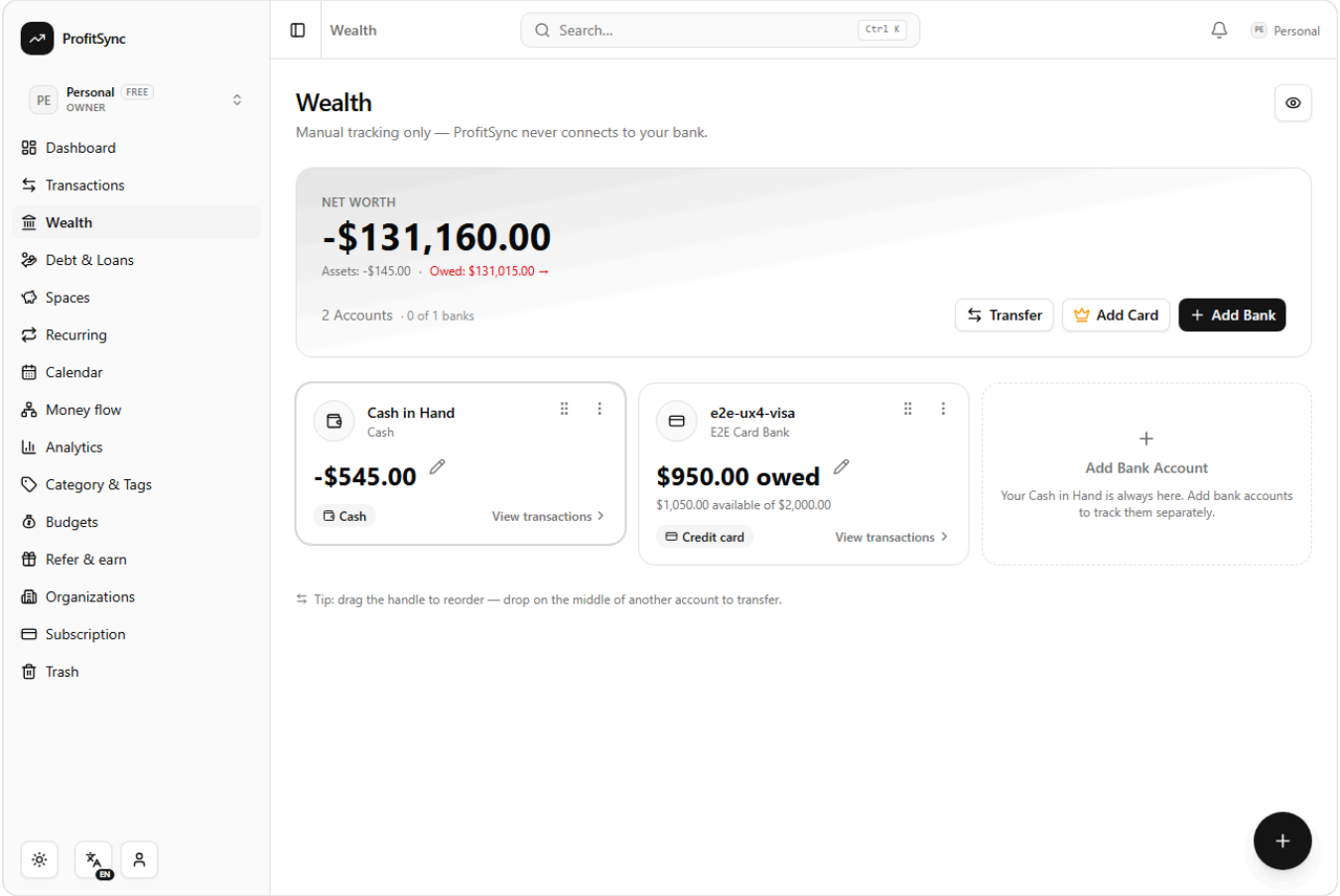
\$306.33

\$238.67

Show full schedule

Save the most interest is estimated to save the most interest.

390 px viewport — the same screens (full-page captures; the fixed bottom tab bar appears mid-image as a capture artifact).



/wealth — net worth now subtracts loans as well as cards and adds receivables; the "Owed" line links to the hub. (Demo data from the e2e run: the payment came out of a €0 cash account.)

6. Integration impact

- **Net worth** (`summarizewealth`): liquid + receivables – liabilities (cards and loans); receivables are assets but never liquid. Same-currency only — no FX is invented.
- **Credit cards** stay a liability in the same position; a loan and a card never double-count (ledger test).
- **Transactions / budgets / analytics**: interest and fees are ordinary categorised expenses, so budgets and reports see them; principal is a transfer and stays out of income / expense through the shared `tx-sql` rules.
- **Trash / restore**: a payment group reverses and re-applies as one, using the group-wide restore fixed in PR #365.
- **Navigation**: "Debt & Loans" in the sidebar, mobile layout and command palette; `debts` `i18n` namespace registered; English base strings with English placeholders in the seven other locales (repo convention, `i18n:check` green).
- **Native shells**: `cap:sync:android` and `build:ios` + `cap copy ios` run after the final gate.

7. Tests

Suite	What it proves	Result
<code>debt-math.test.ts</code>	Periodic rates, annuity payment, amortization convergence and final-payment tolerance, split calculation, date stepping across month ends.	pass
<code>debt-planner.test.ts</code>	Strategy ordering, rollover, minimum-only never rolls over, lump sum and extra effects, stabilisation when under-budgeted, zero-rate assumption flag.	pass
<code>debt-status.test.ts</code>	Derived statuses (overdue, due soon, paid off, lifecycle overrides), month obligations, upcoming window, debt-free estimate and its "unknown" reasons, per-currency totals, insights.	pass
<code>debt-ledger.test.ts</code>	Borrowing is not income; split payment moves bank –500 / loan +420 / expense 80; informal principal-only; delete + restore once; card + loan no double count; receivable flows; multi-currency stays separate.	pass
<code>e2e/debts.spec.ts</code> (local Postgres, Playwright)	Empty state → quick informal debt · borrowing into cash (cash +5,000, debt 5,000, income 0) · schedule + debt-free date · known split (bank –500, principal –420, interest 70 + fees 10 the only expenses) · auto split by rate · delete a payment reverses everything, Trash restore brings it back once · hub tabs · owed-to-me lend/repay with no income or expense · net worth on /wealth · paid off + close with the cash account exactly where the payments left it.	11 / 11
<code>e2e/credit-card.spec.ts</code> · <code>e2e/smoke.spec.ts</code>	Regression after the debts changes.	10 / 10 · 9 / 9
Full unit suite	68 files	793 / 793

Gate: secret scan · check-esm-extensions (235 files) · boot-functions (every function boots under Node ESM) · check-route-guards (126 handlers) · i18n:check · lint · typecheck · vitest run · npm run build + chunk-cycle grep (empty) — all green. Migration 0063 applied to the local Docker database only; the shared Neon database was never touched.

Fixed along the way

The credit-card e2e spec called the API without `x-org-id` and relied on the server's profile fallback; that fallback is served from a 60-second per-user cache, so right after a workspace switch it could still name the previous workspace and cleanup ran against the wrong org. Both specs now pin the workspace id returned by the switch. The "Owed on cards" assertion was updated to the new "Owed" line.

8. Known limitations and follow-ups

Deliberately deferred: refinancing comparison UI (the data model supports it via `refinanced_into_account_id`) · calendar and planned-transaction integration · variable-rate modelling (rates are simulated as fixed) · FX for multi-currency totals (per-currency figures only) · milestones beyond progress % · AI quick add cannot split interest on a loan payment · human translations for the new `debts` keys.

- A typed split is trusted user input; the server checks only that the components sum to the total.
- The org-resolution cache described above affects any client code that omits `x-org-id` after a switch (pre-existing).

Recommendation: internal-test ready. The accounting core and planner are tested unit-level and end to end on a real database. Before production: real translations, a read-through of the planner copy by a finance-literate reviewer, and a staging run of migration 0063.